12-3 Programming Basics [2] (Python)

Point!

1 Loop Structure

(1) Using a ([1]**for statement**) makes it possible to execute a process repeatedly.
A "for" statement is written as shown below.

```
for␣variable␣in␣range([Variable range]):
␣␣␣␣[Process to repeat]
```

In a "for" statement, the indented block of code is executed repeatedly.

(2) In a "for" statement, the range() instruction can be used to specify a range for a variable.

Writing style	Meaning
range(end value)	The value of the variable increases by 1, starting from ([2]**0**) up to the ([3]**end value - 1**).
range(start value, end value)	The value of the variable increases by 1, starting from the ([4]**start value**) up to the ([5]**end value - 1**).
range(start value, end value, increment)	The value of the variable increases by the specified ([6]**increment**), starting from the ([7]**start value**) up to the ([8]**end value - increment**).

<Example> A program that displays integers from 0 to 3

```
01 for i in range(0, 4):      Repeat variable "i" increasing by 1 from 0 to 3.
02     print(i)               Displays the value of variable "i".
```

Execution result

```
0
1
2
3
```

<Example> A program that displays odd numbers from 1 to 5

```
01 for i in range(1, 7, 2):   Repeat variable "i" increasing by 2 from 1 to 6.
02     print(i)               Displays the value of variable "i".
```

Execution result

```
1
3
5
```

2 Branching Structure

(1) ([9]**Comparison operator**) : Operator used to compare expressions or values.

Comparison operator	Meaning	Example	Meaning of the example
==	([10]**Equal**)	x == 70	x is equal to 70
! =	([11]**Not equal**)	x ! = 70	x is not equal to 70
<	([12]**Less than**)	x < 70	x is less than 70
>	([13]**Greater than**)	x > 70	x is greater than 70
<=	([14]**Less than or equal to**)	x <= 70	x is less than or equal to 70
>=	([15]**Greater than or equal to**)	x >= 70	x is greater than or equal to 70

(2) Using ([16]**if statements**) allows conditions to be branched and processed as conditional expressions.

• ([17]**Conditional expression**) : Expression that determines whether conditions are met and returns "true" if met and "false" if not.

[1] Using ([18]**if ~ else**) makes it possible to describe the process that occurs when a condition is not met.

```
if␣[Conditional expression]:
    [Process when a condition is true]
else:
    [Process when a condition is false]
```

<Example> Program for determining pass or fail

01	`x = 70`	Assign the value `70` to variable `x`.
02	`if␣x >= 60:`	If `x` is `60` or greater,
03	`␣␣␣␣result = 'Pass'`	If true, assign "`Pass`" to the variable `result`.
04	`else:`	If not,
05	`␣␣␣␣result = 'Fail'`	Assign "`Fail`" to the variable `result`.
06	`print(result)`	Displays the value of the variable `result`.

Execution result

```
Pass
```

[2] Using ([19]**if ~ elif ~ else**) makes it possible to determine multiple conditions sequentially.

```
if [Conditional expression 1]:
␣␣␣␣[Process when conditional expression 1 is true]
elif [Conditional expression 2]:
␣␣␣␣[Process when conditional expression 1 is false and conditional expression 2 is true]
else:
␣␣␣␣[Process when both conditional expression 1 and conditional expression 2 are false]
```

<Example> Program for evaluating based on test scores

01	`x = 70`	Assign the value `70` to variable `x`.
02	`if␣x >= 90:`	If `x` is `90` or greater,
03	`␣␣␣␣result = 'Grade is A'`	If true, assign "`Grade is A`" to the variable `result`.
04	`elif␣x >= 50:`	Otherwise, if `x` is `50` or greater,
05	`␣␣␣␣result = 'Grade is B'`	Assign "`Grade is B`" to the variable `result`.
06	`else:`	If none of the conditions are met,
07	`␣␣␣␣result = 'Grade is C'`	Assign "`Grade is C`" to the variable `result`.
08	`print(result)`	Displays the value of the variable `result`.

Execution result `Grade is B`

Warm Up

Answer the following questions.

(1) Choose the program that displays "Grade is A" from the options **A** to **D** below, and answer using the letter.

A
```
point = 90
if point >= 85:
    result = 'Grade is A'
else:
    result = 'Grade is B'
```

B
```
point = 90
if point >= 85:
    result = 'Grade is A'
else:
    result = 'Grade is B'
print(result)
```

C
```
point 90
if point >= 85:
    result = 'Grade is A'
else:
    result = 'Grade is B'
print(result)
```

D
```
point 90
if point >= 85:
    result = 'Grade is A'
else:
    result = 'Grade is B'
```

(2) The following program displays the total and the average of all integers from 1 to 10. Fill in the blanks **A** to **C** with the appropriate characters or numbers to complete the program.

```
total = 0
for i in range(1, [ A ]):
    total = [ B ]
average = [ C ] /10
print(total)
print(average)
```

Explanation

(1) "=" is needed to assign a value to the variable "point", and the print function is needed to display the value of the variable "result". Therefore, **B**

(2) Set the variable "total" to an initial value of 0 and then calculate the total by continuously adding integers 1 to 10 as variable "i" to the variable "total". When counting variables from 1 to 10, be sure to specify up to the next number, 11. Therefore, **A**: 11, **B**: total+i, **C**: total

Try

Answer the following questions.

(1) Choose the appropriate execution result when the following program is executed from the options **A** to **D** below, and answer using the letter.

```
for i in range(0, 5, 1):
    print(i)
```

A
```
0
1
2
3
4
```

B
```
0
1
2
3
4
5
```

C
```
1
2
3
4
```

D
```
1
2
3
4
5
```

(2) Choose the program that displays "`Pass`" from the options **A** to **D** below, and answer using the letter.

A
```
score 95
if score < 90:
    result = 'Fail'
else:
    result = 'Pass'
print(result)
```

B
```
score 95
if score < 90:
    result = 'Fail'
else:
    result = 'Pass'
```

C
```
score = 95
if score < 90:
    result = 'Fail'
else:
    result = 'Pass'
```

D
```
score = 95
if score < 90:
    result = 'Fail'
else:
    result = 'Pass'
print(result)
```

(3) The following program counts down from 5 to 0, and when it reaches 0, it displays "`Start!`" Fill in the blanks **A** to **C** with the appropriate characters or numbers to complete the program.

```
for i in [ A ]:
    count = [ B ]
    if [ C ]:
        print('Start!')
    else:
        print(count)
```

(4) The following program displays "`It is an even number`" for even numbers from 1 to 100. Fill in the blanks **A** and **B** with the appropriate characters or numbers to complete the program.

```
for i in range(1, [ A ]):
    if [ B ]:
        print('It is an even number')
    else:
        print(i)
```

Exercise

1 Answer the following questions.

(1) Choose the program that would produce the same execution result as the following program from the options **A** to **D** below, and answer using the letter.

```
for i in range(3):
    print(i)
```

A
```
for i in range(0, 3):
    print(3)
```

B
```
for i in range(0, 3, 1):
    print(i)
```

C
```
for i in range(0, 3, 1):
    print(3)
```

D
```
for i in range(0, 3, 2):
    print(i)
```

(2) Choose the appropriate execution result when the following program is executed from the options **A** to **C** below, and answer using the letter.

```
x = 7
if x < 3:
    print('We will seat you at the counter')
elif x <= 10:
    print('We will seat you at a table')
else:
    print('No seats are available')
```

A `We will seat you at the counter`
B `We will seat you at a table`
C `No seats are available`

(3) The following program displays the total for all numbers from 1 to 100. Fill in the blanks **A** and **B** with the appropriate characters or numbers to complete the program.

```
total = 0
for i in range(1, [ A ]):
    total = [ B ]
print(total)
```

(4) The following program displays the number of even numbers between 1 and 10. Fill in the blanks **A** to **C** with the appropriate characters or numbers to complete the program.

```
for i in [ A ]:
    if i % 2 == 0:
        count = count + [ B ]
print( [ C ] )
```